BO YANG

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RESEARCH FOCUS

Generative models of multi-level human mobility (individual, household, population): diffusion, LLMs, constrained generation.

EDUCATION

University of California, Los Angeles, Ph.D. in Civil Engineering (Transportation) — Sep. 2025 – Present
National University of Singapore, Master of Computing in Computer Science (CAP: 4.15/5) — Jan. 2022 – Jul. 2023
Nanjing University, B.Eng. in Industrial Engineering (GPA: 4.36/5) — Sep. 2017 – Jul. 2021

RESEARCH EXPERIENCE

UCLA Mobility Lab, Graduate Student Researcher (Advisor: Prof. Jiaqi Ma) — Sep. 2025 – Present

- **Diffusion model for whole-day activity generation**: Designed a conditional diffusion model with a Transformer denoiser that jointly generates *location* × *activity* × *time* for a full day (5,697 zones × 15 activities × 96 time slots); trained on 8.1M GPS-derived person-days from 618K individuals in Los Angeles County.
- Beat 8 generative baselines (incl. MIRAGE, TrajGPT, MoveSim, ActVAE, TimeGeo) on 5 of 6 held-out realism metrics, reducing activity-duration and chain-structure divergence by 45% and 44% vs. the strongest baseline.
- **Constrained generation**: Developed maximum-entropy guidance that adapts the generator to a new target year from published aggregate totals alone, matching fine-tuning on individual-level data on the constrained statistics and cutting distance error in unseen regions from 14.1 to 1.1 km. Ongoing: counterfactual generation for large-event scenarios.
- **Trip purpose inference [1], co-first author**: Proposed a weakly supervised, uncertainty-aware framework that combines POI semantic zones with Pareto calibration to travel-survey statistics, reducing activity-type, start-time, and duration divergence by 23%, 48%, and 12% on 81M GPS staypoints.
- **LLM-to-CVAE mobility generation [3]**: Led population synthesis (households reconstructed from census aggregates) and location assignment for PopTraj, which distills an LLM teacher into a CVAE to generate 395K household-coordinated person-weeks in Tokyo (39% lower divergence than the best of 9 baselines).
- **Tourist mobility [2]**: Contributed to LLM-based activity-chain generation conditioned on seasonal spatial priors.

MIT-NEU-UF Research Program, Advisors: Prof. Jinhua Zhao & Dr. Yuebing Liang — May 2024 – Jun. 2025

- **Deep RL for urban planning [4], first author**: Formulated transit-oriented development planning as sequential decision-making with a combinatorial action space (land use × floor-area ratio) and a multi-objective reward; improved spatial efficiency by 70% and 54% over rule-based and optimization baselines.

NUS Master's Capstone, vision-language pipeline for dashcam driving-style profiling — Feb. 2023 – May 2023
NJU Undergraduate Thesis, deep RL with a Transformer policy for TSP local search — Jan. 2021 – Jun. 2021

PUBLICATIONS (* EQUAL CONTRIBUTION)

[1] **Yang, B.***, Ma, H.*, Liu, Y., Zhang, Z., Stanford, C., Sun, M., & Ma, J. Uncertainty-Aware Trip Purpose Inference from GPS Trajectories via POI Semantic Zones and Pareto Calibration. *IEEE ITSC*, 2026. [arXiv]

[2] Liu, Y.*, Sang, Y.*, Liao, X., Sun, M., **Yang, B.**, Zhang, Z., Stanford, C., Ma, H., & Ma, J. GPS-Enhanced Tourist Mobility Modeling with Seasonal Spatial Priors and LLM-Based Activity Chain Generation. *IEEE ITSC*, 2026. [arXiv]

[3] Ma, H., Liu, Y., **Yang, B.**, Zhang, Y., Stanford, C., Fox, E., Liao, X., & Ma, J. PopTraj: Hierarchical Multi-Source Generation of Household-Coordinated Weekly Activity–Location Trajectories with Urban Population Synthesis. *Information Fusion*, under review (2nd round). [SSRN]

[4] **Yang, B.**, Liang, Y., Zhao, Z., Ding, F., Zheng, Y., Wang, S., & Zhao, J. Performing Transit-Oriented Development Planning Using Deep Reinforcement Learning. *Transportation Research Part C*, under review (2nd round).

ADDITIONAL EXPERIENCE

Temasek Labs @ NUS, Graduate Researcher: ANN vector search for radar-signal retrieval — Jun. 2022 – Sep. 2022
NUS Smart Medical Tech Lab, Research Intern: vision-based vital-signs detection — Sep. 2021 – Dec. 2021

SERVICE & SKILLS

Service: Editorial Coordinator, IEEE Open Journal of ITS (OJ-ITS); Student Committee Member, IEEE ITSS Technical Committee on Trustworthy AI
ML: Diffusion models, VAEs/CVAEs, Transformers, LLM prompting and distillation, deep RL, constrained/guided generation
Programming: Python, PyTorch, SQL; also TensorFlow, MATLAB, R, Java, C